```
/**
 * server/lib/ffmpeg-config.ts
 *
 * Fix #2 — FFmpeg Blocked Container for 90 Minutes
 *
 * Centralized FFmpeg encoding constraints for Replit's CPU-only environment.
 * Import these args wherever you spawn FFmpeg — StreamEditor, pre-encoder,
 * shorts-clip-publisher. Never let FFmpeg run without these limits on Replit.
 *
 * Key changes:
 *   - preset: ultrafast (software x264, no hardware accel on Replit)
 *   - scale: cap at 720p (Shorts don't need 1080p, saves ~60% encode time)
 *   - hard timeout: 10 minutes max (was 90 minutes — kills runaway jobs fast)
 *   - Shorts duration cap: 60 seconds enforced at encode time
 */

export const FFMPEG_SHORTS_ARGS = [
  '-vf',      'scale=-2:720',       // cap at 720p, preserve aspect ratio
  '-c:v',     'libx264',
  '-preset',  'ultrafast',          // fastest encode — Replit has no GPU
  '-crf',     '28',                 // acceptable quality at ultrafast
  '-maxrate', '2500k',
  '-bufsize',  '5000k',
  '-c:a',     'aac',
  '-b:a',     '128k',
  '-t',       '60',                 // hard cap: YouTube Shorts max = 60 seconds
  '-movflags', '+faststart',        // optimize for streaming playback
] as const;

export const FFMPEG_LONGFORM_ARGS = [
  '-vf',      'scale=-2:720',
  '-c:v',     'libx264',
  '-preset',  'ultrafast',
  '-crf',     '26',                 // slightly better quality for long-form
  '-maxrate', '3000k',
  '-bufsize',  '6000k',
  '-c:a',     'aac',
  '-b:a',     '128k',
  '-movflags', '+faststart',
  // No -t cap for long-form — duration is intentional
] as const;

/** Hard timeout in milliseconds. Kill any FFmpeg job that exceeds this. */
export const FFMPEG_TIMEOUT_MS = 10 * 60 * 1000; // 10 minutes

/**
 * Wraps an FFmpeg spawn with the hard timeout.
 * Replace your existing FFmpeg process manager call with this.
```

```
 *
 * Usage:
 *   const result = await spawnFfmpegWithTimeout(inputPath, outputPath, FFMPEG_SHORTS_ARGS);
 */
export async function spawnFfmpegWithTimeout(
  inputPath:  string,
  outputPath: string,
  extraArgs:  readonly string[] = FFMPEG_SHORTS_ARGS,
): Promise<void> {
  const { spawn } = await import('child_process');
  const { createLogger } = await import('./logger.js');
  const log = createLogger('ffmpeg');

  return new Promise((resolve, reject) => {
    const args = [
      '-i', inputPath,
      ...extraArgs,
      '-y',          // overwrite output without prompting
      outputPath,
    ];

    log.debug(`[FFmpeg] Spawning: ffmpeg ${args.join(' ')}`);

    const proc = spawn('ffmpeg', args, { stdio: ['ignore', 'pipe', 'pipe'] });

    // Hard timeout — kill if exceeded
    const killTimer = setTimeout(() => {
      proc.kill('SIGKILL');
      reject(new Error(`FFmpeg hard timeout — encoding exceeded ${FFMPEG_TIMEOUT_MS / 60000}
minutes, killed`));
    }, FFMPEG_TIMEOUT_MS);

    proc.on('close', code => {
      clearTimeout(killTimer);
      if (code === 0) {
        resolve();
      } else {
        reject(new Error(`FFmpeg exited with code ${code}`));
      }
    });

    proc.on('error', err => {
      clearTimeout(killTimer);
      reject(err);
    });
  });
}
```